

Q1. Define Covariance and explain how it differs from Correlation in terms of scale and interpretation.

Ans:

Covariance

- Measures how two variables **move together**
- Positive → both increase/decrease together
- Negative → one increases, other decreases
- Value depends on **units and scale** (not standardized)
- Hard to interpret magnitude directly

Correlation

- Standardized form of covariance
- Measures both **direction and strength** of relationship
- Value always between **-1 and +1**
- **Unit-free** (independent of scale)
- Easy to interpret and compare across datasets

Key Differences

- Covariance → **scale-dependent**, correlation → **scale-free**
- Covariance → shows only direction, correlation → shows **direction + strength**
- Correlation is more **useful for comparison and interpretation**

Q2. What does a positive, negative, and zero covariance indicate about the relationship between two variables?

Ans:

Positive Covariance

- Both variables move in the same direction
- If one increases, the other also increases (and vice versa)
- Indicates a direct relationship

Negative Covariance

- Variables move in opposite directions
- If one increases, the other decreases
- Indicates an inverse relationship

Zero Covariance

- No linear relationship between the variables
- Changes in one variable do not predict changes in the other

Q3. Discuss the limitations of covariance as a measure of relationship between two variables. Why is correlation preferred in many cases?

Ans:

Limitations of Covariance

- Depends on units (scale-dependent): Value changes if units change (e.g., meters → cm), so it's not consistent
- Hard to interpret magnitude: A value like 200 or 2000 doesn't clearly tell how strong the relationship is
- Not bounded: Can take any value ($-\infty$ to $+\infty$), so comparisons across datasets are difficult
- Only shows direction: Tells whether the relationship is positive or negative, but not how strong it is clearly
- Sensitive to data scale: Larger values in data can inflate covariance, making it misleading

Why Correlation is Preferred

- Standardized measure: Always lies between -1 and $+1$
- Unit-free: Not affected by scale or units
- Easy to interpret: Clearly shows both direction and strength
- Comparable across datasets: Can compare relationships across different variables easily

Q4. Explain the difference between Pearson's correlation coefficient and Spearman's rank correlation coefficient. When would you prefer to use Spearman's correlation?

Ans:

Pearson's Correlation Coefficient

- Measures **linear relationship** between two variables
- Uses **actual numerical values**
- Assumes data is **normally distributed**
- Sensitive to **outliers**
- Best for **continuous data with linear trends**

Spearman's Rank Correlation Coefficient

- Measures **monotonic relationship** (not necessarily linear)
- Uses **ranks instead of actual values**
- Does **not require normal distribution**
- Less sensitive to **outliers**
- Works well with **ordinal or ranked data**

Key Differences

- Pearson → uses **raw data**, Spearman → uses **ranked data**
- Pearson → **linear only**, Spearman → **monotonic (increasing/decreasing)**
- Pearson → affected by outliers, Spearman → more robust

When to Prefer Spearman's Correlation

- When data is **not normally distributed**
- When relationship is **not linear but still increasing/decreasing**
- When data contains **outliers**
- When working with **ranked or ordinal data** (e.g., rankings, ratings)

In short:

Use **Pearson** for linear, clean data; use **Spearman** when data is messy, ranked, or non-linear

Q5. If the correlation coefficient between two variables X and Y is 0.85, interpret this value in context. Can you infer causation from this value? Why or why not?

Ans:

Interpretation of $r = 0.85$

- Indicates a strong positive relationship between X and Y
- As X increases, Y also tends to increase
- The value is close to +1, so the relationship is strong and consistent, though not perfect

Can we infer causation?

- No, correlation does not imply causation

Why not?

- There may be a third variable (confounding factor) affecting both X and Y
- The relationship could be coincidental
- We don't know the direction of influence ($X \rightarrow Y$ or $Y \rightarrow X$)
- Correlation only shows association, not cause-effect

Example:

Ice cream sales and drowning incidents may have high correlation, but both are influenced by summer weather, not each other.

Q6. Using the dataset below, calculate the covariance between X and Y.

X	2	4	6	8
Y	3	7	5	10

Ans:

We calculate covariance using the formula:

$$Cov(X, Y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n}$$

First find the means:

- Mean of X = $(2 + 4 + 6 + 8) / 4 = 5$
- Mean of Y = $(3 + 7 + 5 + 10) / 4 = 6.25$

Now compute deviations and multiply:

- $(2-5)(3-6.25) = (-3)(-3.25) = 9.75$
- $(4-5)(7-6.25) = (-1)(0.75) = -0.75$
- $(6-5)(5-6.25) = (1)(-1.25) = -1.25$
- $(8-5)(10-6.25) = (3)(3.75) = 11.25$

Sum = $9.75 - 0.75 - 1.25 + 11.25 = 19$

Now divide by $n = 4$:

$$\text{Cov}(X, Y) = 19 / 4 = 4.75$$

Q7. Compute the Pearson correlation coefficient between variables A and B:

A	10	20	30	40	50
B	8	14	18	24	28

Ans:

We compute Pearson correlation using:

$$r = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{\sum(x-\bar{x})^2 \cdot \sum(y-\bar{y})^2}}$$

First find means:

- Mean of A = $(10+20+30+40+50)/5 = 30$
- Mean of B = $(8+14+18+24+28)/5 = 18.4$

Now compute deviations and sums:

- $\sum(A-\bar{A})(B-\bar{B}) = 500$
- $\sum(A-\bar{A})^2 = 1000$
- $\sum(B-\bar{B})^2 = 252.8$

Now substitute:

$$r = 500 / \sqrt{(1000 \times 252.8)}$$

$$r = 500 / \sqrt{252800}$$

$$r = 500 / 502.79 \approx 0.995$$

Q8. The following table shows heights (in cm) and weights (in kg) of 5 students. Find the correlation coefficient between Height and Weight.

Height	150	160	165	170	180
Weight	50	55	58	62	70

Ans:

We compute Pearson correlation using:

$$r = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{\sum(x-\bar{x})^2 \cdot \sum(y-\bar{y})^2}}$$

First find means:

- Mean Height = $(150 + 160 + 165 + 170 + 180) / 5 = 165$
- Mean Weight = $(50 + 55 + 58 + 62 + 70) / 5 = 59$

Now compute values:

- $\sum(x-\bar{x})(y-\bar{y}) = 250$
- $\sum(x-\bar{x})^2 = 500$
- $\sum(y-\bar{y})^2 = 238$

Substitute:

$$r = 250 / \sqrt{500 \times 238}$$

$$r = 250 / \sqrt{119000}$$

$$r \approx 250 / 344.96 \approx 0.725$$

Q9. Given the dataset below, determine whether there is a positive or negative correlation between X and Y. (No need for exact calculation, just reasoning.)

X	1	2	3	4	5
Y	15	12	9	7	3

Ans: Negative correlation

Reason (simple):

As X increases ($1 \rightarrow 5$), the values of Y decrease ($15 \rightarrow 3$).

This means both variables move in opposite directions, which indicates a negative relationship.

Q10. Two investment portfolios have the following returns (%) over 5 years. Compute the covariance and correlation coefficient, and interpret whether the portfolios move together.

Year	Portfolio A	Portfolio B
1	8	6
2	10	9
3	12	11
4	9	8
5	11	10

Ans:

First find means:

- Mean of Portfolio A = $(8 + 10 + 12 + 9 + 11) / 5 = 10$
- Mean of Portfolio B = $(6 + 9 + 11 + 8 + 10) / 5 = 8.8$

♦ **Covariance formula**

$$Cov(X, Y) = \frac{\sum(x - \bar{x})(y - \bar{y})}{n}$$

After calculation:

$$Cov(A, B) = 2.8$$

Correlation formula

$$r = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sqrt{\sum(x - \bar{x})^2 \cdot \sum(y - \bar{y})^2}}$$

After calculation:

$r \approx 0.96$

- Covariance = 2.8
- Correlation ≈ 0.96

Summary:

- Covariance is positive, so both portfolios move in the same direction
- Correlation is very high (close to +1)